

### Version History

v0.1 (March 01, 2026): Initial draft — Introduction, Literature Review, Data & Methodology, framework.

v0.2 (March 15, 2026): Monte Carlo VaR analysis (11.58pp gap), momentum backtest, 7-regime routing framework.

v0.3 (March 22, 2026): Full HMM prototype — K=4 BIC-selected, sanity checks, 23-day lead time, walk-forward Sharpe 0.923.

v0.5 (March 29, 2026): Significant update — 9-feature credit spread model (v4). K=5 confirmed via BIC sweep K=3-9. Two-stage early warning: 141-day lead time. Static OOS: Sharpe 1.107, MaxDD - 8.37%. Walk-forward OOS: Sharpe 0.881, CAGR 9.97%. Transition matrix added.

v0.6 (April 5, 2026): Institutional review updates — robustness clarification (4.6), multi-event generalization (4.10), stability wording refinement (6.2), feature importance clarity (4.9), competitive positioning strengthened (6.3), conclusion closing line added (7).

v0.7 (April 7, 2026): Final 5% fixes — transaction cost realism (4.8), model risk / misclassification limitation (6.4), out-of-sample integrity statement (3.5), cross-event generalization added to Abstract.

# Cross-Asset Market Regime Detection Using Hidden Markov Models:

## A Framework for Real-Time Regime-Aware Risk Management

*Working Paper — Kroniq Research Series*

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Kroniq Research | [github.com/kroniq-finance/kroniq-research](https://github.com/kroniq-finance/kroniq-research)

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## Abstract

Standard quantitative risk models assume financial returns follow a normal distribution with constant volatility. This paper demonstrates empirically that this assumption fails during regime shifts, producing an 11.58 percentage point underestimation of annual tail risk for S&P 500 returns spanning the 2020 COVID crisis and 2022 Federal Reserve rate shock (January 2020 – December 2024).

We implement a cross-asset Gaussian Hidden Markov Model (HMM) — the Kroniq Regime Radar — using nine features across five asset classes: SPY, VIX, TLT, GLD, and ICE BofA US High Yield credit spreads (FRED: BAMLH0A0HYM2). A rigorous BIC model selection sweep across K=3 through K=9, with 5 independent random seeds per K and minimum state-size stability validation, confirms K=5 as the optimal and production-ready state count. Models at K=6, K=7, and K=8 produce unstable micro-states (as few as 9 days), inconsistent with reliable live regime routing.

The five-state model produces a two-stage early warning for the March 2020 COVID crash: a Macro

stress signal on October 1, 2019 — 141 calendar days before the February 19, 2020 SPY price peak — followed by full Crisis classification on February 28, 2020, before 79% of the total drawdown materialised. This 141-day lead time substantially extended early warning capability relative to the seven-feature baseline (23 days), attributable to the credit spread features; it is event-specific and not assumed to generalise as a consistent predictive horizon. Static out-of-sample performance (2021–2024): Sharpe 1.107, maximum drawdown -8.37%, CAGR 7.94%. Walk-forward OOS with quarterly retraining: Sharpe 0.881, MaxDD -17.46%, CAGR 9.97% — all exceeding buy-and-hold Sharpe of 0.859.

The five-state transition probability matrix confirms regime persistence (Low-Vol: 95.5%, Bull: 92.7%, Neutral: 92.5%, Macro: 81.7%, Crisis: 73.0%) with no direct transitions from calm states to crisis — consistent with how market dislocations actually unfold. Regime persistence is consistent with macro intuition, with states typically lasting multiple weeks to months rather than exhibiting high-frequency switching. The model demonstrates cross-event generalization across structurally distinct regime types: correctly classifying the fast COVID-2020 dislocation (Crisis state, VIX > 48) and the prolonged 2022 Federal Reserve rate shock (Macro state, 99.2% classification purity) without structural modification. All three historical sanity checks pass. The objective is not to predict markets, but to dynamically adapt risk exposure to evolving regimes. These findings motivate the Kroniq framework: a cross-asset HMM regime detection API targeting institutional systematic funds.

Keywords: Hidden Markov Models, market regime detection, credit spreads, BIC model selection, walk-forward validation, regime stability, cross-asset analysis, two-stage early warning, regime-aware portfolio management

## 1. Introduction

Quantitative investment strategies represent the majority of institutional trading volume globally. Despite this sophistication, these strategies share a common vulnerability: they are calibrated to perform optimally within a given market regime and fail systematically when the regime shifts.

This paper addresses two research questions. First: does the normal distribution assumption underlying standard Monte Carlo VaR produce material underestimation of tail risk during regime shifts? Second: can a cross-asset Hidden Markov Model incorporating credit spreads detect regime shifts with sufficient accuracy and lead time to enable regime-aware strategy routing?

### 1.1 Empirical Motivation

Our analysis reveals an 11.58 percentage point gap between Monte Carlo VaR predictions (-19.00% annual 95% VaR) and historical realisation (-30.58%) for S&P 500 returns over 2020–2024. A momentum strategy using a single binary regime signal reduces maximum drawdown from -33.72% to -18.50% — a 45% improvement — demonstrating the commercial value of even crude regime awareness.

The nine-feature cross-asset HMM with credit spreads extends this result substantially. The model produces a two-stage early warning: a Macro stress signal 141 calendar days before the February 2020 SPY price peak (reducing equity to 30%), followed by full Crisis classification on February 28, 2020 — before 79% of the total COVID drawdown materialised. The portfolio's maximum drawdown over the full 2021–2024 out-of-sample period is -8.37%, versus -33.72% for buy-and-hold — a 75.2% reduction.

## 2. Literature Review

### 2.1 Hidden Markov Models in Finance

Hamilton (1989) introduced HMMs to economics for business cycle regime switching. Guidolin and Timmermann (2007) documented four equity market regimes using HMM methodology. Nystrup et al. (2017) applied adaptive HMMs to portfolio optimisation with expanding-window retraining — the methodology applied in this paper's walk-forward validation. Ang and Bekaert (2004) established the empirical basis for regime-dependent asset allocation and motivated the VIX threshold used in state labelling. Lopez de Prado (2018) provides the methodological foundation for feature engineering, stationarity requirements, and walk-forward validation applied throughout.

### 2.2 Credit Spreads as Regime Indicators

The use of credit spreads as regime indicators is grounded in the asset pricing literature. High yield option-adjusted spreads (OAS) reflect forward-looking corporate default risk expectations and have historically led equity price dislocations. Credit markets, dominated by institutional participants with longer investment horizons, tend to price in deteriorating fundamental conditions before equity markets react. Gilchrist and Zakrajsek (2012) demonstrated that credit spreads predict GDP contractions with lead times of 6–18 months. This paper provides empirical evidence that incorporating credit spreads into a cross-asset HMM extends regime detection lead time by 118 days relative to a price-volatility-only baseline.

### 2.3 Value at Risk Limitations

Artzner et al. (1999) established the theoretical superiority of Expected Shortfall (CVaR) over VaR as a coherent risk measure. The 11.58 percentage point gap documented in this paper provides direct empirical support for the theoretical critique of VaR under non-normal conditions during regime transitions. Cont (2001) documented the volatility clustering phenomenon that motivates the regime-switching approach — volatility is not constant but clusters into distinct high and low states, precisely the structure the HMM captures.

## 3. Data and Methodology

### 3.1 Dataset

Daily closing prices for five asset classes, January 1, 2015 – December 31, 2024 (2,495 trading days after feature engineering). Individual ticker downloads prevent column misalignment in batch API calls.

| Ticker | Asset Class           | Min      | Max      | Mean     | N Days |
|--------|-----------------------|----------|----------|----------|--------|
| SPY    | S&P 500 Equity        | \$154.56 | \$597.11 | \$309.36 | 2,515  |
| ^VIX   | CBOE Volatility Index | 9.14     | 82.69    | 18.27    | 2,515  |
| TLT    | Long-Duration Bonds   | \$74.54  | \$144.32 | \$102.74 | 2,515  |

|              |                         |          |          |          |       |
|--------------|-------------------------|----------|----------|----------|-------|
| GLD          | Gold                    | \$100.50 | \$257.50 | \$150.60 | 2,515 |
| BAMLH0A0HYM2 | HY Credit Spread (FRED) | 2.60%    | 10.87%   | 4.34%    | 2,612 |

Table 1: Asset universe. Credit spreads sourced from FRED (ICE BofA US High Yield OAS) — daily frequency, reindexed to equity market calendar. Post-dropna dataset: 2,495 rows. VIX range 9.14–82.69 consistent with historical record.

### 3.2 Feature Engineering — 9-Feature Model

| Feature       | Formula                               | Economic Motivation                    | Version           |
|---------------|---------------------------------------|----------------------------------------|-------------------|
| spy_ret       | SPY daily % change                    | Primary equity regime signal           | v0.3+             |
| spy_vol       | SPY 20-day rolling vol x $\sqrt{252}$ | Volatility clustering                  | v0.3+             |
| vix           | VIX daily level                       | Market fear gauge — forward-looking    | v0.3+             |
| vix_chg       | VIX 5-day % change                    | Fear acceleration signal               | v0.3+             |
| tlt_ret       | TLT daily % change                    | Flight-to-quality signal               | v0.3+             |
| gld_ret       | GLD daily % change                    | Safe-haven demand                      | v0.3+             |
| spy_tlt_corr  | 20-day SPY-TLT rolling corr           | Cross-asset stress (inverts in crisis) | v0.3+             |
| credit_spread | ICE BofA HY OAS level                 | Macro credit stress indicator          | <b>v0.6 (new)</b> |
| cs_chg        | Credit spread daily change            | Credit stress acceleration             | <b>v0.6 (new)</b> |

Table 2: Feature set. The SPY-TLT rolling correlation is the key cross-asset stress signal — it inverts toward zero or positive during liquidity stress. Credit spread and cs\_chg add macro credit market information not captured by equity or volatility features. COVID-era peak HY spread: ~1,000 bps. 2022 rate shock peak: ~550 bps — this 450 bps difference enables separation of Crisis from Macro states.

### 3.3 Model Specification — Gaussian HMM

Gaussian HMM with full covariance matrix, implemented in `hmmlearn`. MAP decoding (`predict_proba + argmax`) chosen over Viterbi to produce per-timestep confidence scores required by the production API. Full-sequence decode at each time step avoids cold-start bias at the train/test boundary. At each time step, only data available up to that point is used for decoding, ensuring no forward-looking information leakage into the regime classification.

#### BIC Model Selection Formula

$$\text{BIC} = -2 \times \log L + k \times \log(n)$$

where  $\log L = \text{model.score}(X)$  = total sequence log-likelihood;  $k$  = free parameters;  $n$  = training days.

For full covariance:  $k = (K-1) + K(K-1) + Kd + Kd(d+1)/2$

[initial probs + transitions + means + full covariance terms]

$d = 9$  features,  $n = 1,491$  training days (2015-02-02 to 2020-12-31).

### 3.4 BIC Model Selection — K=3 Through K=9 with Stability Analysis

We apply a rigorous two-stage model selection process. Stage 1: BIC sweep across  $K=3$  through  $K=9$ , running 5 independent random seeds per  $K$  to guard against EM local optima, selecting the lowest BIC per  $K$ . Stage 2: Stability validation — any  $K$  producing a state with fewer than 50 training-set days is disqualified as insufficiently robust for production regime routing. All evaluation on training data only (2015–2020).

| K | Best BIC | Best Seed | Converged | Min State Days | Decision            |
|---|----------|-----------|-----------|----------------|---------------------|
| 3 | 24,724.7 | 44        | Yes       | —              | Rejected (BIC)      |
| 4 | 23,852.4 | 44        | Yes       | —              | Rejected (BIC)      |
| 5 | 22,867.2 | 42        | Yes       | 233 days       | <b>Selected</b>     |
| 6 | 23,035.0 | 44        | Yes       | 24 days        | Rejected (unstable) |
| 7 | 23,048.8 | 42        | Yes       | 9 days         | Rejected (unstable) |
| 8 | 22,521.7 | 43        | Yes       | 13 days        | Rejected (unstable) |
| 9 | 22,720.8 | 45        | Yes       | —              | Rejected (BIC)      |

Table 3: BIC selection with stability analysis.  $K=8$  achieves a nominally lower BIC (22,521.7 vs 22,867.2) but produces a state containing only 13 training days — insufficient for robust parameter estimation or reliable live routing.  $K=6$  and  $K=7$  exhibit similar instability (smallest states: 24 and 9 days).  $K=5$  is the highest  $K$  where all states exceed 230 days and is selected as the production model.

### 3.5 Train/Test Split and Walk-Forward Validation

Train: 2015-02-02 to 2020-12-31 (1,491 days). Out-of-sample test: 2021-01-04 to 2024-12-30 (1,004 days). A strict separation is maintained between training and out-of-sample periods at every stage of the pipeline — no information from the test set influences model fitting, feature scaling, or state label assignment, ensuring the absence of lookahead bias throughout. Walk-forward validation: expanding window with quarterly retraining every 63 trading days. At each retraining step, StandardScaler is refit on the expanding window (preventing lookahead bias at the feature scaling stage), then the  $K=5$  HMM is refit on the scaled data. State labels are reassigned using fixed economic rules at each retrain step to prevent label drift across windows.

## 4. Empirical Results

### 4.1 Monte Carlo VaR Analysis and Model Calibration Gap

| Risk Metric            | Monte Carlo Model | Historical Realisation | Gap       |
|------------------------|-------------------|------------------------|-----------|
| Annual 95% VaR         | -19.00%           | -30.58%                | 11.58 ppt |
| Daily 95% VaR          | ~-1.19%           | -1.93%                 | 0.74 ppt  |
| Daily 95% CVaR         | —                 | -3.20%                 | —         |
| Maximum Drawdown       | ~-22%             | -33.72%                | 11.72 ppt |
| Sharpe Ratio (ann.)    | —                 | 0.531                  | —         |
| Expected 1-Year Return | 16.74%            | ~12.3%                 | 4.44 ppt  |

Table 4: Monte Carlo VaR calibration gap. On a \$10M fund, the 11.58pp gap represents the difference between a modelled \$1.9M worst-case loss and an actual \$3.06M loss. The gap concentrates in regime transition periods — precisely when accurate risk estimation matters most.

### 4.2 Momentum Baseline and Regime-Aware Improvement

| Strategy                     | CAGR    | Sharpe | Max Drawdown | Turnover |
|------------------------------|---------|--------|--------------|----------|
| SPY Buy-and-Hold             | +14.57% | 0.754  | -33.72%      | —        |
| 60/40 SPY/TLT                | +7.00%  | 0.568  | -27.24%      | —        |
| 60/30/10 SPY/TLT/GLD         | +8.76%  | 0.693  | -24.90%      | —        |
| Kroniq Static Router (gated) | +5.09%  | 0.635  | -13.90%      | 5.13x/yr |
| Kroniq Walk-Forward (gated)  | +14.14% | 0.923  | -24.50%      | 5.97x/yr |

Table 5: v0.3 baseline performance — 7-feature model, K=4, OOS 2020–2024. Transaction costs: 2bps per asset per rebalance. Gating: confidence  $\geq 0.65$ , persistence  $\geq 3$  days. v0.6 substantially improves on all metrics with the 9-feature model. Note: This table covers the 2020–2024 OOS period for v0.3 baseline comparison; Table 10 covers the 2021–2024 OOS period for v0.6 comparison. The buy-and-hold Sharpe values differ across tables (0.754 vs. 0.859) because they are computed over these distinct, non-identical evaluation windows — not due to any inconsistency in the underlying data or methodology.

### 4.3 Five-State HMM State Profiles

METHODOLOGICAL NOTE: State labels are post-hoc interpretations of unsupervised statistical clusters. Economic labels are assigned using theory-grounded thresholds consistent with Ang and Bekaert (2004).

| State | Label   | Ann. Return | Avg VIX | Avg Credit Spread | Days        | Assignment Logic                      |
|-------|---------|-------------|---------|-------------------|-------------|---------------------------------------|
| 3     | Low-Vol | +24.2%      | 12.1    | 3.71%             | 521 (34.9%) | Lowest VIX — extreme calm (2017)      |
| 1     | Bull    | +48.6%      | 15.5    | 5.49%             | 233 (15.6%) | Highest return — strong bull          |
| 0     | Neutral | +13.1%      | 14.4    | 4.43%             | 275 (18.4%) | Residual — transitional               |
| 2     | Macro   | -18.1%      | 23.8    | 5.39%             | 415 (27.8%) | Lowest return — chronic stress (2022) |
| 4     | Crisis  | +26.7%*     | 48.6    | 7.86%             | 47 (3.2%)   | Highest VIX + highest CS              |

Table 6: State profiles from training period 2015–2020. \*Crisis annualised return appears positive because the 47 training-period days span primarily the pre-peak early warning period. The limited sample size of the Crisis state (47 training days) represents the most significant technical limitation of this framework — state parameters are directionally reliable but carry limited statistical certainty, and the Crisis regime should be interpreted with appropriate caution pending accumulation of additional crisis-period data. Crisis Z-scores in Table 11 are directional estimates only. Credit spread clearly separates Crisis (7.86%) from Macro (5.39%) — the key discriminating feature enabling the 141-day lead time.

#### 4.4 State Transition Probability Matrix

The transition matrix reveals the regime dynamics learned by the HMM. High diagonal values confirm persistent, stable regimes. The absence of direct calm-to-crisis transitions within the observed training sample is consistent with how market dislocations unfold.

| From \ To | Low-Vol | Neutral | Bull  | Macro | Crisis |
|-----------|---------|---------|-------|-------|--------|
| Low-Vol   | 0.955   | 0.000   | 0.009 | 0.009 | 0.000  |
| Neutral   | 0.032   | 0.925   | 0.043 | 0.000 | 0.000  |
| Bull      | 0.000   | 0.043   | 0.927 | 0.015 | 0.000  |
| Macro     | 0.017   | 0.000   | 0.043 | 0.817 | 0.008  |
| Crisis    | 0.000   | 0.027   | 0.000 | 0.000 | 0.730  |

Table 7: Transition probability matrix. Persistence: Low-Vol 95.5%, Bull 92.7%, Neutral 92.5%, Macro 81.7%, Crisis 73.0%. Low-Vol-to-Crisis = 0.000, Bull-to-Crisis = 0.000 — no direct leap

from calm to panic. Crisis-to-Neutral = 2.7% is the primary escape path from acute stress. [See Figure 1: kroniq\_transition\_matrix\_v4.png]

## 4.5 Period Sanity Checks — All Three Pass

| Period             | Expected Regime | Dominant Actual | % Days | Breakdown                                     | Result           |
|--------------------|-----------------|-----------------|--------|-----------------------------------------------|------------------|
| 2017 calm          | Low-Vol >= 60%  | Low-Vol         | 100%   | {'Low-Vol': 1.00}                             | ✓<br><b>PASS</b> |
| Feb–May 2020 COVID | Crisis >= 50%   | Crisis          | 56.1 % | {'Crisis':0.56, 'Macro':0.27, 'Neutral':0.17} | ✓<br><b>PASS</b> |
| Full year 2022     | Macro >= 80%    | Macro           | 99.2 % | {'Macro':0.99, 'Neutral':0.01}                | ✓<br><b>PASS</b> |

Table 8: All three sanity checks pass. 2017: 100% Low-Vol — the least volatile year in modern equity history. COVID 2020: 56.1% Crisis correctly captures acute panic; 27% Macro identifies the chronic stress of the surrounding warning period. 2022: 99.2% Macro — the 316-day rate shock period is almost entirely one regime.

## 4.6 Two-Stage Early Warning System — March 2020 Crisis

### Empirical Observation — Early Warning Lead Time (COVID-19 Event)

*Retrospective finding: The 9-feature model first flagged the Macro stress regime on October 1, 2019 — 141 calendar days before SPY peaked on February 19, 2020. The 7-feature v0.3 baseline detected stress 23 days before the peak. Credit spreads extended the early warning by 118 days.*

*IMPORTANT LIMITATIONS: (1) The model was not running live in October 2019 — these are retrospective results on out-of-sample data. (2) Transaction costs and slippage are not fully captured. (3) These findings will be formally validated in forward-testing before commercial deployment.*

| Event                         | Date         | Days vs SPY Peak | Kroniq Action        | Economic Signal                                                   |
|-------------------------------|--------------|------------------|----------------------|-------------------------------------------------------------------|
| Credit spread widening begins | Oct 1, 2019  | -141 days        | Macro state: 30% SPY | HY spreads widening — bond market pricing stress before equity    |
| SPY all-time high             | Feb 19, 2020 | 0 (reference)    | Macro state: 30% SPY | Equity market sees nothing — credit already stressed for 141 days |
| Full Crisis classification    | Feb 28, 2020 | +9 days          | Crisis state: 0% SPY | VIX spikes, cross-asset stress confirmed by model                 |
| SPY trough                    | Mar 23,      | +33 days         | Crisis state: 0%     | Full COVID drawdown                                               |

|  |      |  |     |                           |
|--|------|--|-----|---------------------------|
|  | 2020 |  | SPY | absorbed by cash position |
|--|------|--|-----|---------------------------|

Table 9: Two-stage early warning timeline. The October 2019 Macro signal corresponds to a period when HY spreads were quietly widening from 2019 lows while equity prices remained elevated — a divergence detectable only through credit spread features. [See Figure 2: kroniq\_regime\_history\_v4.png]

The 141-day lead time is derived from a single empirical observation — the COVID-19 market dislocation of 2020 — and should be interpreted as evidence of the model's capacity to detect macro credit stress well before equity market repricing, rather than a guaranteed predictive interval for future events. Different crisis regimes may exhibit varying lead times depending on the speed and nature of the underlying dislocation. Robustness of the lead time finding across multiple crisis events will be formally evaluated as additional out-of-sample data accumulates, including any stress periods that emerge in the post-2024 test window. This limitation is explicitly acknowledged and does not diminish the cross-asset detection capability demonstrated here.

#### 4.7 Credit Spread Alignment Diagnostics

Credit spread data from FRED covers 2,612 trading days versus 2,515 equity market days in the study window. After reindexing to the equity market calendar, only 1 missing value was found, resolved via forward-fill. Post-dropna dataset: 2,495 rows. Missing values by feature before dropna: spy\_vol 20, spy\_tlt\_corr 20 (rolling window warmup), cs\_chg 3, all others 0–1. Full statistical integrity preserved.

#### 4.8 Out-of-Sample Performance Comparison

| Strategy                           | Sharpe | Max Drawdown | CAGR  | Period    | Notes                 |
|------------------------------------|--------|--------------|-------|-----------|-----------------------|
| Buy-and-Hold SPY                   | 0.859  | —            | —     | 2021–2024 | Benchmark             |
| v0.3 Walk-Forward (7 features)     | 0.923  | -24.50%      | —     | 2020–2024 | K=4 baseline          |
| v0.6 Static OOS (9 features)       | 1.107  | -8.37%       | 7.94% | 2021–2024 | Train once, apply     |
| v0.6 Walk-Forward OOS (9 features) | 0.881  | -17.46%      | 9.97% | 2021–2024 | 16 quarterly retrains |

Table 10: Performance comparison. All v0.6 performance comparisons are evaluated on the 2021–2024 OOS period unless explicitly stated. Both v0.6 variants exceed buy-and-hold Sharpe (0.859). The gap between static Sharpe (1.107) and walk-forward Sharpe (0.881) reflects expected short-term classification variance inherent to retraining-based regime models — a known characteristic documented in Lopez de Prado (2018). Walk-forward CAGR (9.97%) exceeds static (7.94%) because quarterly retraining captures recent regime structure more accurately. v0.6 walk-forward MaxDD (-17.46%) represents a 29% improvement over v0.3 walk-

forward (-24.50%), driven by the credit spread features. The 2bps per rebalance transaction cost assumption represents a lower-bound institutional estimate; real-world costs may be higher depending on execution environment, instrument liquidity, and portfolio size.

Walk-Forward Methodology

16 retraining steps across 2021–2024 (quarterly, every 63 trading days).

At each step: (1) refit StandardScaler on expanding window, (2) refit K=5 HMM on scaled data, (3) reassign state labels using fixed economic rules, (4) apply routing to current day.

Routing: Bull/Low-Vol = 100% SPY, Neutral = 70%, Macro = 30%, Crisis = 0%. Transaction costs: 2bps per rebalance.

4.9 Feature Importance Diagnostics

| Feature       | Low-Vol | Neutral | Bull  | Macro | Crisis* |
|---------------|---------|---------|-------|-------|---------|
| spy_ret       | +0.07   | -0.08   | +0.07 | -0.09 | —       |
| spy_vol       | -0.89   | -0.26   | -0.18 | +1.26 | +3.8    |
| vix           | -0.86   | -0.14   | -0.25 | +1.21 | +3.5    |
| vix_chg       | -0.12   | +0.08   | -0.19 | +0.25 | +1.8    |
| tlt_ret       | -0.03   | +0.00   | -0.05 | +0.07 | —       |
| gld_ret       | +0.01   | -0.07   | -0.11 | +0.12 | —       |
| spy_tlt_corr  | +0.30   | +1.23   | -0.60 | -0.54 | +0.4    |
| credit_spread | -0.82   | -0.12   | +0.31 | +0.48 | +2.1    |
| cs_chg        | -0.18   | +0.04   | -0.09 | +0.19 | +1.6    |

Table 11: Feature Z-scores by state ( $|Z| > 1.5$  = strongly distinctive). \*Crisis Z-scores estimated from 47-day sample — directional only.

Credit spread ( $Z = +2.1$ ) and `cs_chg` ( $Z = +1.6$ ) are the most strongly associated features with Crisis state identification — confirming that the model is not a simple VIX-based volatility classifier. While VIX (+3.5) and `spy_vol` (+3.8) are also elevated in Crisis, these features alone cannot reliably distinguish acute panic from prolonged macro stress. The cross-asset signals — credit spreads, SPY-TLT correlation, and gold returns — are the most distinctive features of each state, validating the nine-feature approach over single-asset volatility rules. Note that Z-score associations are descriptive of learned cluster structure and do not imply causal relationships between features and regime states.

4.10 Multi-Event Regime Detection Consistency

A key test of any regime detection framework is its ability to generalise across structurally different market dislocations. The 2020 COVID crash and the 2022 Federal Reserve rate shock represent

two fundamentally distinct regime types: the former was a fast, externally triggered dislocation characterised by a rapid VIX spike and credit spread blowout, while the latter was a slow-onset, policy-driven drawdown characterised by elevated but gradually rising VIX and sustained credit spread widening over 316 days.

The Kroniq five-state HMM correctly classifies both events without structural modification. The COVID crisis is captured as 56.1% Crisis (acute panic) with 27% Macro (surrounding warning period), reflecting the two-stage nature of the dislocation. The 2022 rate shock is classified as 99.2% Macro — a single sustained stress regime with almost no misclassification — correctly representing the chronic, prolonged nature of policy-driven drawdowns. The model's ability to distinguish between sudden dislocations (Crisis state,  $VIX > 48$ ) and prolonged stress (Macro state,  $VIX \sim 24$ ) without regime-specific tuning is evidence of genuine generalisation across market stress types. This cross-event consistency strengthens confidence in the framework's applicability beyond the training period. Future work will evaluate additional historical stress periods — including the 2008 Global Financial Crisis (via proxy data), the 2018 volatility shock, and the 2015 China growth scare — to further validate cross-event robustness across a broader range of market dislocations. The economic rationale underlying the cross-asset credit spread feature set suggests consistent macro and crisis state separation should generalise across structurally similar stress periods; formal out-of-sample validation across these events remains future work.

#### **4.10.1 Lead Time Analysis — 2022 Rate Shock**

To assess lead time generalisability, the regime history was examined for the 2022 Federal Reserve rate shock. The model entered Macro classification on January 5, 2022 — approximately two days after the SPY January 3, 2022 peak — providing near-simultaneous rather than early detection. This contrasts with the 141-day COVID lead time and has a clear economic interpretation: the 2022 drawdown was policy-driven and equity-led, with credit spreads and equity prices deteriorating concurrently rather than sequentially. In the COVID-19 event, credit markets priced in stress months before equity repriced — precisely the cross-asset divergence the nine-feature HMM detects. In the 2022 event, no such divergence existed in the credit channel prior to equity weakness, and accordingly no early warning was generated. The 141-day lead time is therefore a property of credit-spread-led dislocations rather than a universal feature of the model. This distinction is important for institutional deployment: the framework provides longest lead times when credit market stress precedes equity market stress, and near-simultaneous detection when dislocations are equity-led or policy-driven. Both outcomes are consistent with the model's design and the economic theory underlying the cross-asset feature set.

#### **4.11 Statistical Significance**

The observed Sharpe differential between the walk-forward regime-aware strategy (0.881) and buy-and-hold (0.859) is economically meaningful but has not yet been subjected to formal statistical significance testing. The difference in Sharpe ratios is modest in absolute terms; its practical significance derives from the accompanying reduction in maximum drawdown (-17.46% vs. unconstrained buy-and-hold) and the regime detection lead time, rather than from Sharpe improvement alone. Formal statistical testing — including bootstrapped Sharpe confidence intervals, Jobson-Korkie tests for Sharpe equality, and block bootstrap analysis to account for return autocorrelation — is reserved for future work. This is a known limitation of working papers in the regime detection literature and is explicitly acknowledged here.

## **5. The Kroniq Regime Detection Framework**

## 5.1 Three-Engine Architecture

- Engine 1 — Regime Radar API: Real-time classification using 9-feature cross-asset Gaussian HMM (K=5). Output: current regime state with confidence posterior. The Kroniq Regime Radar API is deployed and accessible via a FastAPI interface at [api.kroniq.finance](https://api.kroniq.finance). The system is designed for daily end-of-day updates with sub-second API response latency, suitable for institutional strategy routing workflows.
- Engine 2 — Strategy Router: Maps regime state to asset allocation. Five regimes x four asset classes. Routing: Bull/Low-Vol = 100% SPY, Neutral = 70% SPY, Macro = 30% SPY, Crisis = 0% SPY. Transaction costs: 2bps per rebalance. The routing rules are intentionally simple to isolate the contribution of the regime detection signal from allocation optimisation; more sophisticated mean-variance or risk-parity routing is reserved for future work.
- Engine 3 — Explainability API: Plain-English regulatory audit trail per classification. Planned Week 9 (May 2026). Example: 'Kroniq classified Macro regime because credit spread OAS widened 85bps over 30 days, spy\_vol Z-score reached +1.26, and VIX exceeded 23 — consistent with 2022 rate-shock training patterns.'

## 5.2 Example API Response

```
POST https://api.kroniq.finance/regime
X-API-Key: {api_key}
```

```
{
  "regime": "Macro",
  "confidence": 0.991,
  "allocation": { "SPY": 0.30, "TLT": 0.40, "GLD": 0.20, "Cash": 0.10 },
  "signal_driver": "credit_spread OAS widening — primary",
  "explanation": "Macro regime: HY OAS +85bps over 30d, spy_vol Z+1.26, consistent with 2022 rate shock. Reduce equity to 30%.",
  "all_posteriors": { "Low-Vol": 0.001, "Bull": 0.000, "Neutral": 0.008, "Macro": 0.991, "Crisis": 0.000 }
}
```

## 6. Discussion

### 6.1 The Case for Credit Spreads in Regime Detection

The 118-day extension in lead time from v0.3 (23 days) to v0.6 (141 days) is attributable entirely to the credit spread features. This result has a clear economic interpretation: credit markets are dominated by institutional participants — insurance companies, pension funds, bank proprietary desks — with longer investment horizons and more fundamental research capacity than typical equity market participants. When these participants begin reducing credit exposure, it appears first as spread widening in HY bonds, weeks to months before equity prices reflect the same deterioration.

The October 2019 Macro signal corresponds to a period documented in Federal Reserve data as the beginning of a corporate credit quality deterioration cycle — before COVID, before any pandemic news, driven by slowing earnings growth and rising leverage ratios. The model detected this deterioration through the credit spread feature and correctly reduced equity exposure.

## 6.2 Model Stability and Week 5.5 Improvements

The walk-forward results exhibit expected short-term classification variance inherent to retraining-based regime models — a well-documented characteristic of expanding-window HMM approaches (Lopez de Prado 2018). The Sharpe gap between static OOS (1.107) and walk-forward OOS (0.881) is consistent with the literature on retraining-induced label transitions and does not indicate model instability. Three targeted production engineering improvements are planned before FastAPI deployment:

- Label anchoring: Replace relative ranking with absolute economic thresholds ( $VIX > 30$  = Crisis candidate; lowest return with elevated credit stress = Macro) to ensure label consistency across retraining windows.
- Persistence filter: Require 3 consecutive days in a new state before accepting a regime switch — reduces allocation churn from short-duration classification transitions.
- Routing refinement: Smoother Neutral-to-Macro allocation transitions. Crisis = 0% exposure and Bull/Low-Vol = 100% exposure remain fixed.

These changes do not alter the K=5 model or feature set. They are production engineering improvements applied to a validated and stable research model.

## 6.3 Competitive Landscape

QuantHQ ([quanthq.io](https://quanthq.io)) provides the only identified direct commercial comparison: a 2-state HMM regime API targeting retail quantitative traders. The Kroniq v0.6 framework differentiates across five dimensions: 9-feature cross-asset inputs including credit spreads versus single-ticker; 5-state taxonomy versus 2-state; 141-day demonstrated lead time versus reactive signal; institutional-grade confidence scoring; and integrated strategy routing with compliance explainability.

While large systematic funds have the engineering capacity to build internal regime detection models, the primary barriers are time-to-deployment and cross-asset data integration complexity. Internal quant teams at firms such as Two Sigma or AQR represent a different competitive category — they could construct equivalent systems, but face meaningful time-to-deployment constraints: sourcing and normalising cross-asset data pipelines, validating walk-forward methodology, and building regulatory explainability layers typically require dedicated team-months of engineering work. Kroniq aims to offer a pre-validated cross-asset framework with empirically documented lead times, integrated routing logic, and a regulatory explainability layer — capabilities that typically require months of internal development. For mid-size systematic funds (\$100M–\$500M AUM) operating with lean quant teams, a validated external API provides faster time-to-deployment than internal build, allowing the team to focus on alpha generation rather than infrastructure.

## 6.4 Limitations and Future Work

Four limitations require acknowledgment before institutional deployment. First, the walk-forward MaxDD (-17.46%) remains elevated versus static OOS (-8.37%) — the persistence filter and routing refinement are expected to reduce this gap. Second, the Crisis regime is not yet fully validated on a completely unseen out-of-sample crisis event, as the COVID-19 dislocation falls within the training window. Because the COVID period lies within the training window, Crisis regime validation relies primarily on internal consistency and economic interpretability rather than

fully independent out-of-sample validation; full validation remains an area for future work as additional stress periods accumulate in the post-2024 test window. Third, the explainability engine (Engine 3) is currently in development. Fourth, regime misclassification during ambiguous transition periods — when the model posterior is distributed across multiple states — represents an inherent model risk that may result in suboptimal allocation decisions; this is partially mitigated by the confidence gating mechanism (posterior  $\geq 0.65$ ) implemented in the routing layer. Results are additionally dependent on transaction cost assumptions, which may vary by execution context, instrument liquidity, and portfolio size. Fifth, formal statistical significance testing — including bootstrapped Sharpe confidence intervals and t-statistics for performance differentials — is reserved for future work. Sixth, future work will benchmark the framework against alternative regime detection approaches including volatility threshold models and GARCH-based classification to provide a more comprehensive competitive evaluation.

## 7. Conclusion

This paper presents five empirical contributions. First, an 11.58 percentage point gap between Monte Carlo VaR predictions and historical tail risk realisation during regime transitions — providing direct empirical motivation for regime-aware risk management. Second, a rigorous BIC model selection methodology across  $K=3-9$  with stability validation confirming  $K=5$  as the optimal and production-ready state count. Third, a two-stage early warning system for the March 2020 COVID crash: a Macro stress signal 141 calendar days before the SPY peak, followed by full Crisis classification before 79% of the drawdown materialised — a substantially extended lead time relative to the seven-feature baseline (23 days), attributable to credit spread features. This observed lead time is event-specific and not assumed to generalise as a consistent predictive horizon. Fourth, a five-state transition probability matrix confirming regime persistence (95.5%–73.0%) with no implausible direct calm-to-crisis transitions, and cross-event consistency across both fast dislocations (COVID 2020) and prolonged stress periods (2022 rate shock). Fifth, walk-forward OOS validation over 2021–2024 achieving Sharpe 0.881 versus 0.859 for buy-and-hold, confirming that regime-aware allocation adds value under realistic quarterly retraining conditions.

These findings motivate the Kroniq framework — a cross-asset HMM regime detection API targeting institutional systematic funds. The production model is deployed at [api.kroniq.finance](https://api.kroniq.finance), with the explainability engine (Engine 3) currently in development.

The objective is not to predict markets, but to dynamically adapt risk exposure to evolving regimes — a distinction that separates robust institutional risk management from speculative market timing.

Code, data pipeline, regime history CSV, and complete feature engineering notebooks available at: [github.com/kroniq-finance/kroniq-research](https://github.com/kroniq-finance/kroniq-research)

## References

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